

STEP 15 — EXTERNAL VALIDATION OF STRANDED CAPACITY

HDI-N project | Testing Φ against demand-side survey evidence | Criterion fixed before the test is run

OBJECTIVE

Stranded capacity Φ has so far been shown to be internally consistent and reproducible. It has not been shown to correspond to anything observed independently of the indices used to build it. This step supplies that test: it asks whether Φ predicts how much citizens actually use digital services, once the level of state capacity is already accounted for.

The source is the World Bank Global Findex Database 2025, a demand-side household survey covering roughly 141 economies. It is independent of the UN E-Government Survey, it is behavioural rather than assessed, and unlike OECD or EU digital-government statistics it covers low-income economies — which is where Φ is largest and where the test has to work.

The interpretive criterion is fixed in advance and appears at the end of this memo. It is stated there so that the outcome cannot be selected after the fact. A null result is a valid outcome and is reported as such.

TASKS

Task	What to do	Deliverable	Gate / verification
1	Retrieve the country-level Global Findex 2025 file from the World Bank Global Findex download page. Take the most recent survey wave available and record which wave that is. Capture three indicators for all economies, adult population, at whatever variable names the file uses: (i) received government payments or transfers into an account; (ii) made or received a digital payment; (iii) used a mobile phone or the internet to access an account.	Step15_Findex_Raw.csv Columns: ISO3, Country_Findex, GovPay_Pct, DigitalPay_Pct, MobileAccess_Pct, Survey_Year, Retrieval_Date. Plus a note giving the exact variable name used for each of the three, as it appears in the source file.	GATE. Report the row count and how many of the 184 canonical economies match on ISO3, before any analysis. Also state whether the government-payments indicator is reported for all adults or only for transfer recipients — the two are different denominators and the distinction changes the interpretation.
2	Merge onto HDIN_Canonical_v2.csv on ISO3. Define the validation set as economies with a computable Φ and at least one Findex indicator present. Do not drop economies for missing high-technology exports or any Tier II or Tier III requirement. This test uses Tier I only.	Step15_Validation_Set.csv One row per economy in the validation set, carrying ISO3, Country, EGDI, OSI, TII, p, Phi_EGDI, Phi_OSI, Signed_Div, I_income, M49_Subregion, Global_North_South, and the three Findex indicators with their survey year.	Report the size of the validation set and the list of canonical economies with no Findex match. State separately how many of the twenty highest-Φ economies are retained, since the test is uninformative if the facade cases are the ones that drop out.
3	Run the pre-specified test, for each of the three Findex indicators in turn, as the dependent variable Y. Model 1 (baseline): $Y = a + b \cdot \text{EGDI} + c \cdot \text{I_income}$ Model 2 (test): $Y = a + b \cdot \text{EGDI} + c \cdot \text{I_income} + d \cdot \text{Phi_EGDI}$ The coefficient d on Φ is the quantity of interest. Use heteroskedasticity-robust HC3 standard errors throughout.	Step15_Validation_Results.md For each indicator: both models, with coefficients, robust standard errors, p-values, N, R-squared, and the change in R-squared from Model 1 to Model 2.	Report all three indicators whatever they show. Do not select the one with the strongest result, and do not add, drop or transform variables to improve fit. The specification above is fixed.
4	Run three pre-specified sensitivity checks on the same models: (i) substitute Phi_OSI for Phi_EGDI; (ii) add a Global North indicator; (iii) restrict to Global	Appended to Step15_Validation_Results.md as a separate section, clearly marked as sensitivity	Report the variance inflation factor for EGDI, I_income and Phi_EGDI in Model 2. Φ is built from EGDI, so collinearity is expected and must be

Task	What to do	Deliverable	Gate / verification
	South economies only. Report a correlation matrix of EGD, Φ , Φ _EGDI, I_{income} and the three Findex indicators.	rather than the main test. Plus the correlation matrix as a table.	quantified rather than assumed away.
5	Produce one diagnostic scatter per indicator: the Model 1 residual on the vertical axis against Φ _EGDI on the horizontal, with a fitted line. No styling work required — these are for inspection, not for the paper.	Three PNG files at 150 dpi, named Step15_Residual_<indicator>.png. Label the ten highest- Φ economies on each.	State whether the relationship looks linear, and name any economy that sits far off the fitted line in either direction.

INTERPRETIVE CRITERION — FIXED BEFORE THE TEST IS RUN

The coefficient d on Φ in Model 2 is the test. The criterion below is set now and is not revised after seeing the result.

- If d is negative and significant at the 5% level for the government-payments indicator, the measure is externally validated: economies with more stranded capacity show lower citizen uptake than their level of state capacity predicts.
- If d is negative and significant for one or both of the two broader indicators but not for government payments, that is partial validation, and is reported as such rather than as validation.
- If d is not significant for any of the three, Φ is not externally validated on this evidence. The paper then reports the null result and treats the absence of an external referent as a stated limitation.
- If d is positive and significant, the measure behaves opposite to its interpretation, and the framework's substantive claim requires reconsideration before submission.

WATCH-OUTS

- The specification is fixed. Do not add controls, drop economies, transform variables, or try alternative indicators in search of a significant coefficient. If something about the specification appears wrong, report it and stop — do not fix it silently.
- Findex survey years may not align with the 2023 reference year used elsewhere. Record the year per economy and report the range. Do not adjust, interpolate or reweight for the mismatch.
- Φ is constructed from EGD, so EGD and Φ are correlated by construction. This is the reason Model 2 controls for EGD rather than omitting it, and the reason Task 4 requires the variance inflation factors. Do not drop EGD to make the Φ coefficient larger.
- Findex indicators are survey estimates with sampling error that varies by economy and indicator. Where the source publishes standard errors or margins of error, capture them; do not treat the point estimates as exact.
- A null result is a legitimate outcome of this step and is reported plainly. It is more useful now than after submission.
- Merge on ISO3 only. Findex country naming differs from UN DESA and ITU conventions.

SEQUENCING

Task 1 is a hard gate: the match count and the denominator question are returned and acknowledged before any merging or modelling begins, because a low match rate among the high- Φ economies would make the test uninformative and the design would need revisiting. Task 2 follows Task 1. Tasks 3 and 4 follow Task 2 and are the same piece of work. Task 5 follows Task 3 and can be done by a second student.

Return the two data files, the results report and the three diagnostic figures together, with the exception of the Task 1 gate figures, which come first and on their own.